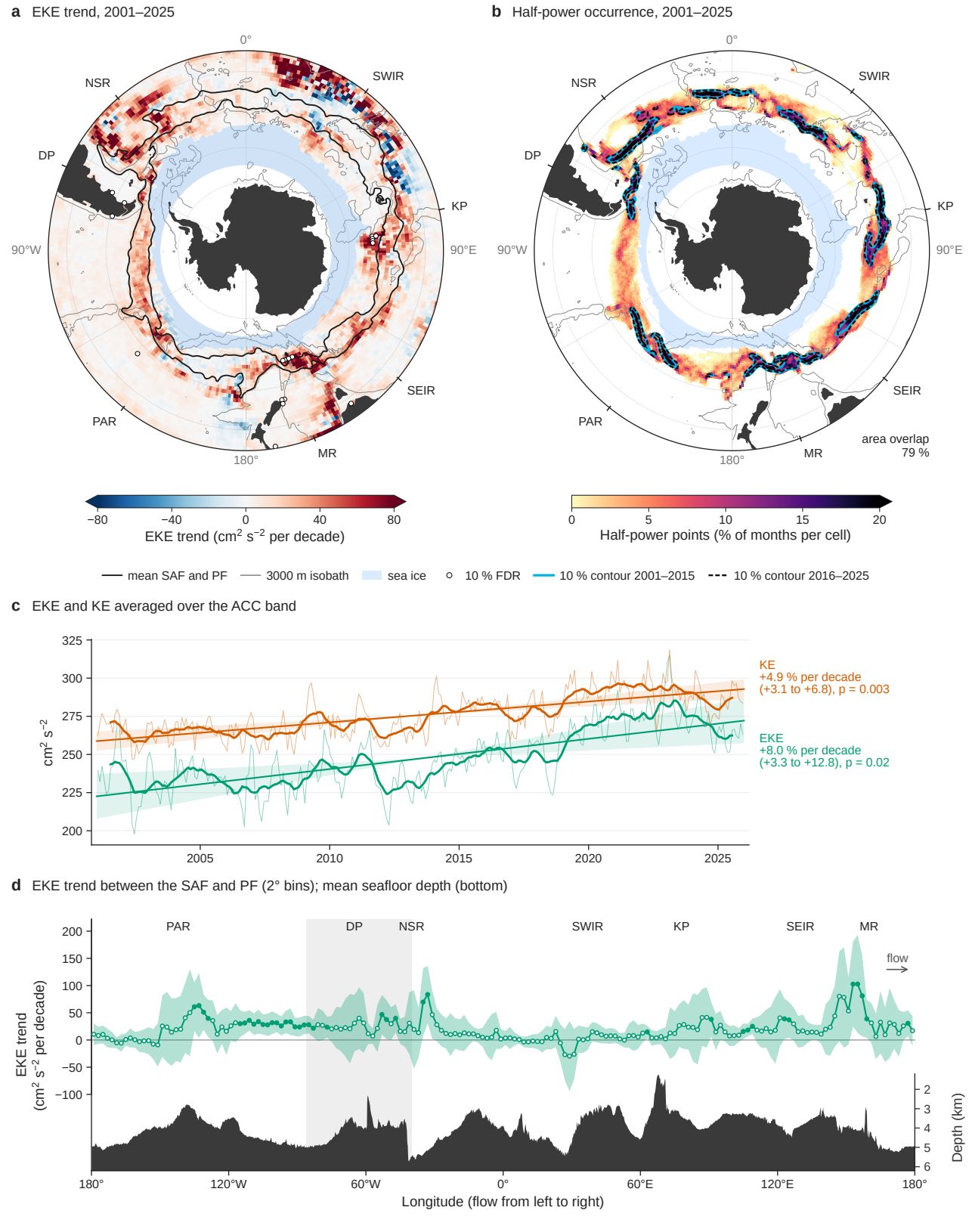

a EKE trend, 2001–2025
b Half-power occurrence, 2001–2025
0°
NSR
SWIR
DP
KP
90°W
90°E
SEIR
PAR
MR
180°
area overlap
79 %
−80
−40
0
40
80
EKE trend (cm² s⁻² per decade)
0
5
10
15
20
Half-power points (% of months per cell)
mean SAF and PF
3000 m isobath
sea ice
10 % FDR
10 % contour 2001–2015
10 % contour 2016–2025
c EKE and KE averaged over the ACC band
cm² s⁻²
325
300
275
250
225
200
2005
2010
2015
2020
2025
KE
+4.9 % per decade
(+3.1 to +6.8), p = 0.003
EKE
+8.0 % per decade
(+3.3 to +12.8), p = 0.02
d EKE trend between the SAF and PF (2° bins); mean seafloor depth (bottom)
PAR
DP
NSR
SWIR
KP
SEIR
MR
flow
EKE trend
(cm² s⁻² per decade)
200
150
100
50
0
−50
−100
Depth (km)
2
3
4
5
6
180°
120°W
60°W
0°
60°E
120°E
180°
Longitude (flow from left to right)